

W3-PM-FINAL

PASSWORD CRACKING REPORT

JOHN THE RIPPER & NETWORKWALKS TOOLS

CYBERSECURITY | NETWORKWALKS

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Program / Batch	B083D – Networkwalks Cybersecurity Internship
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Modules completed	W3-PM1 (Password Cracking with JTR) W3-PM2 (Password Cracking with Networkwalks Tools)
Target files	My-Locked-PDF1.pdf, My-Locked-PDF2.pdf, My-Locked-PDF3.pdf
Phases covered	Password Hash Extraction & Offline Password Cracking

1. Liability Disclaimer

I have performed these activities only on the systems and files provided as part of the Networkwalks Cybersecurity Internship. All materials are for education and research purposes only. Do not use any of these techniques against systems or files you do not own or do not have explicit written permission to test. Unauthorized password cracking is illegal in most jurisdictions.

2. Introduction

This report documents the Week 3 practical modules of the Networkwalks Cybersecurity Internship. Two complementary approaches to password cracking were used against password-protected PDF files:

- **W3-PM1** – Offline cracking using John the Ripper (Johnny GUI) on Windows.
- **W3-PM2** – Online cracking using Networkwalks Hash Calculator and Password Cracker tools.

Both methods follow the same high-level process: extract the password hash from the protected PDF, then attempt to recover the plaintext password using a dictionary attack.

3. Tools Used

Tool	Purpose
John the Ripper (Jumbo)	Offline password cracker (CLI engine)
Johnny (GUI)	Graphical front-end for John the Ripper
Networkwalks Hash Calculator	Extract crackable \$pdf\$ hash from locked PDF
Networkwalks Password Cracker	Browser-based dictionary attack against PDF hashes
OnlineHashCrack PDF Extractor	Alternative online tool used to extract PDF hashes for JTR

4. Activities Performed

4.1 Password Cracking with John the Ripper (W3-PM1)

John the Ripper (Jumbo) and its graphical interface Johnny were installed on Windows. Hashes were extracted from the locked PDFs using an online PDF hash extractor, saved as text files, then loaded into Johnny for a dictionary attack.

Results – John the Ripper

PDF File	Cracked Password	Flag Captured
My-Locked-PDF1.pdf	password1	nw{networkwalks_flag1_jtr_270521_1}
My-Locked-PDF2.pdf	password1	nw{networkwalks_persistence_jtr_270521}
My-Locked-PDF3.pdf	1qaz2wsx	nw{networkwalks_flag_260821_1}

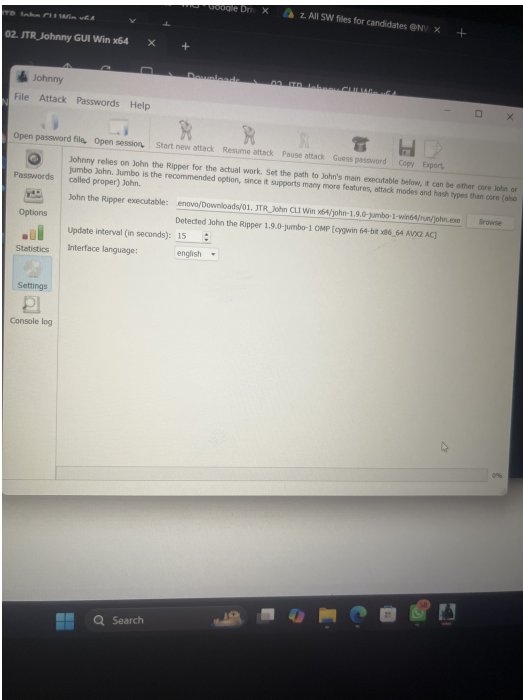


Figure 1: Johnny configured with John the Ripper executable

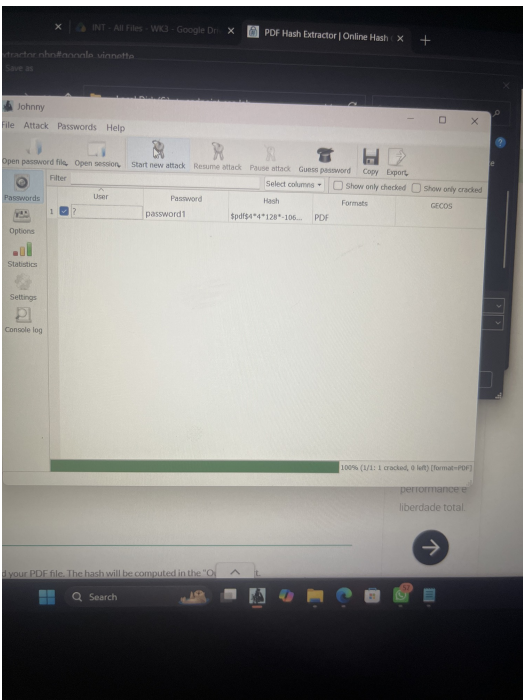


Figure 2: Password 'password1' cracked in Johnny

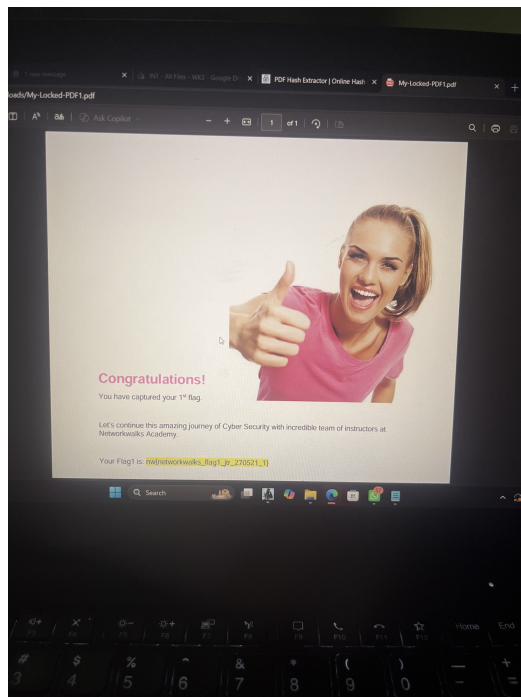


Figure 3: Flag captured from My-Locked-PDF1.pdf

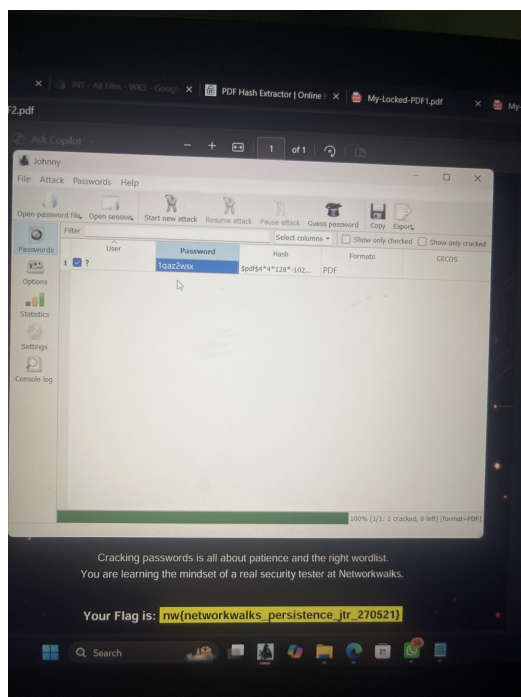


Figure 4: Password '1qaz2wsx' cracked in Johnny

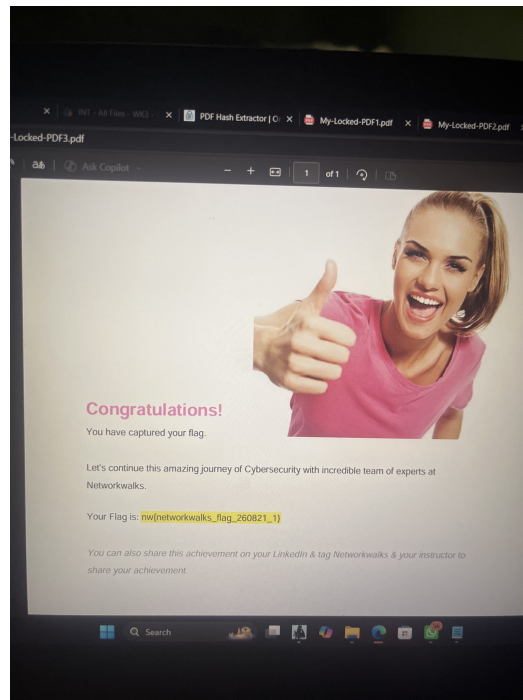


Figure 5: Flag captured from My-Locked-PDF3.pdf

4.2 Password Cracking with Networkwalks Tools (W3-PM2)

The same locked PDFs were processed using the free browser-based tools provided by Networkwalks. The Hash Calculator extracted the \$pdf\$ hash, which was then submitted to the Password Cracker for a dictionary attack.

Results – Networkwalks Tools

PDF File	Cracked Password	Notes
My-Locked-PDF1.pdf	password1	Hash extracted via Networkwalks Hash Calculator (PDF tab)
My-Locked-PDF (additional)	1qaz2wsx	Successfully cracked with Networkwalks Password Cracker

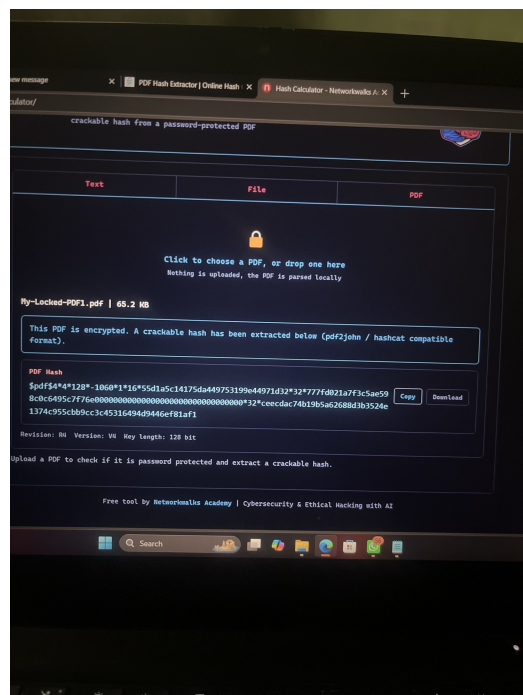


Figure 6: Networkwalks Hash Calculator – PDF hash extracted

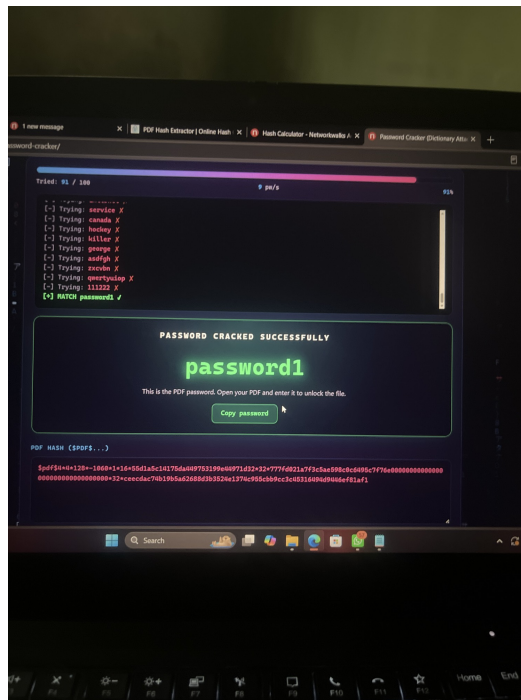


Figure 7: Networkwalks Password Cracker – password1 found

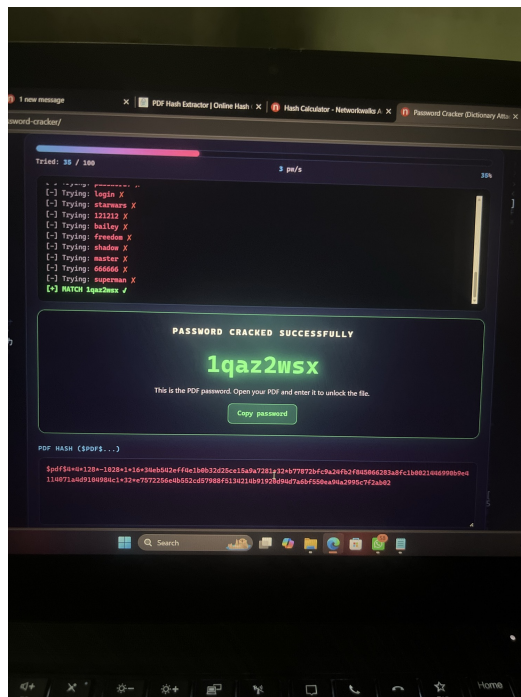


Figure 8: Networkwalks Password Cracker – 1qaz2wsx found

5. Summary of Captured Flags

```
nw{networkwalks_flag1_jtr_270521_1}
nw{networkwalks_persistence_jtr_270521}
nw{networkwalks_flag_260821_1}
```

6. Key Learnings & Recommendations

- Weak and common passwords (password1, 1qaz2wsx) are cracked in seconds with a simple dictionary attack.
- Both offline tools (John the Ripper) and online browser tools can recover the same passwords when the hash is correctly extracted.

- Always use strong, unique passwords (long, mixed character sets) for any password-protected document.
- Password cracking should only be performed on files you own or have explicit written authorization to test.

7. Conclusion

During Week 3 of the Networkwalks Cybersecurity Internship I successfully performed password cracking against multiple protected PDF files using two different toolchains: John the Ripper (Johnny) and the Networkwalks online Hash Calculator + Password Cracker. All target passwords were recovered and the associated flags were captured. The exercise clearly demonstrated why weak passwords are a serious risk and reinforced the importance of using strong authentication practices.

8. Evidence Collected

All screenshots used in this report are stored in the evidences/ folder of the accompanying GitHub repository.

- 01_johnny_settings.jpg
- 02_johnny_cracked_password1.jpg
- 03_flag1_pdf1.jpg
- 04_flag2_pdf2.jpg
- 05_johnny_1qaz2wsx.jpg
- 06_flag3_pdf3.jpg
- 07_nw_hash_calculator.jpg
- 08_nw_cracked_password1.jpg
- 09_nw_cracked_1qaz2wsx.jpg

— End of Report —

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